

Muhammad Umer Bilal

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RESEARCH PROFILE

Cybersecurity researcher and software engineer with an MS in Computer Networks & Security and 6+ years of industry experience building backend and cloud systems. Thesis research applied ensemble machine learning models to real-time low-rate DDoS attack detection. Ongoing hands-on training extends this into application security, cloud security, and DevSecOps. Seeking a PhD position to advance research at the intersection of machine learning and network/application security.

EDUCATION

MS, Computer Networks & Security

National University of Computer & Emerging Sciences (NUCES-FAST), Islamabad, Pakistan | 2021 – 2023

Thesis: *Dynamic Allocation of Window Size for Detection of Low-Rate DDoS Attacks*. Designed an adaptive window-sizing mechanism combined with an ensemble of machine learning classifiers (RNN, MLP, LSTM, Random Forest, Alternating Decision Tree) to detect low-rate DDoS attacks in real time, evaluated on the UNSW-NB15 dataset. Supervisor: Dr. Qaisar Shafi, Associate Professor, Department of Cyber Security.

BS, Computer Science

National University of Computer & Emerging Sciences (NUCES-FAST), Islamabad, Pakistan | 2016 – 2021

RESEARCH & PUBLICATIONS

- *Dynamic Allocation of Window Size for Detection of Low-Rate DDoS Attacks* — MS thesis research; manuscript in preparation for journal submission.
- Presented related work at the International Conference on IT and Industrial Technologies (ICIT), Islamabad, November 2023.

FUTURE RESEARCH INTERESTS

My PhD research interests build directly on my thesis work and extend into the broader security landscape I have been developing through applied practice. I am particularly interested in: (1) machine learning-based anomaly and intrusion detection for evolving network threats, (2) security in cloud-native and containerized environments, (3) application-layer vulnerability detection and automated security testing, and (4) the practical deployment gap between academic detection models and production-grade security systems — an area where my industry background in backend and cloud engineering offers a distinct, practitioner's perspective.

CONFERENCES, SEMINARS & LEADERSHIP

- International Conference on IT and Industrial Technologies (ICIT), Islamabad — attended and presented, November 2023.
- Team Lead, Mobile Application Development Team — coordinated a small development team across multiple production app releases.
- Event Organizer, Nescon (university technology event), National University of Computer & Emerging Sciences, 2022.

RESEARCH & TECHNICAL SKILLS

Machine Learning for Security:	RNN, MLP, LSTM, Random Forest, ensemble/voting classifiers
Network Security:	Intrusion detection, DDoS mitigation, traffic analysis
Application Security:	Web app penetration testing — SQLi, XSS, access control, SSRF, XXE, CORS, JWT attacks (PortSwigger Web Security Academy, Burp Suite)
Cloud & Backend Engineering:	Node.js, NestJS, Express, REST API design, Oracle Cloud, Firebase, PostgreSQL, MySQL
Languages & Tools:	Python, JavaScript/TypeScript, SQL, Bash, MATLAB, Git

PROFESSIONAL EXPERIENCE

Senior Software Engineer — SoftBuilders Software Design LLC, Dubai, UAE

Nov 2024 – Present

Design and build backend systems and APIs for production mobile applications, with a focus on performance, scalability, and secure data handling.

IT Consultant — Emcredit, Dubai, UAE

Aug 2024 – Oct 2024

Audited enterprise IT infrastructure, documented and monitored firewall rules for secure network access, and managed access control and authorization.

Software Engineer — AlphaSquad Technologies / GreenAge Services, Pakistan

2021 – 2024

Developed production applications (Pluckers, Agronomics, Farmacie) with an emphasis on secure backend architecture, API design, and database management.

CERTIFICATIONS & CONTINUING EDUCATION

- Preparing for eJPT (INE Security), progressing toward OSCP
- Extensive hands-on labs — PortSwigger Web Security Academy (SQL injection, XSS, access control, SSRF, XXE, CORS, JWT attacks, file upload RCE)
- Microsoft Azure AZ-900 (in progress)

LANGUAGES

English — full professional proficiency (graduate degree completed entirely in English); Urdu — native

REFERENCES

Dr. Qaisar Shafi — Associate Professor, Department of Cyber Security, NUCES-FAST, Islamabad (MS Thesis Supervisor) | qaisar.shafi@nu.edu.pk

Subhan Ullah, PhD — Assistant Professor, Department of Computer Science, NUCES-FAST, Islamabad (Course Instructor, Advanced Computer Networks) | suban.ullah@nu.edu.pk

Hina Binte Haq — Lecturer, FAST School of Computing, NUCES-FAST, Islamabad (Course Instructor, Applied Information Security) | hina.haq@nu.edu.pk